

Learned social interaction history prioritizes face perception and attentional selection

Qingsong Tan, Ke Jia, Mengyuan Gong

Department of Psychology and Behavioral Sciences, Zhejiang University, 310058



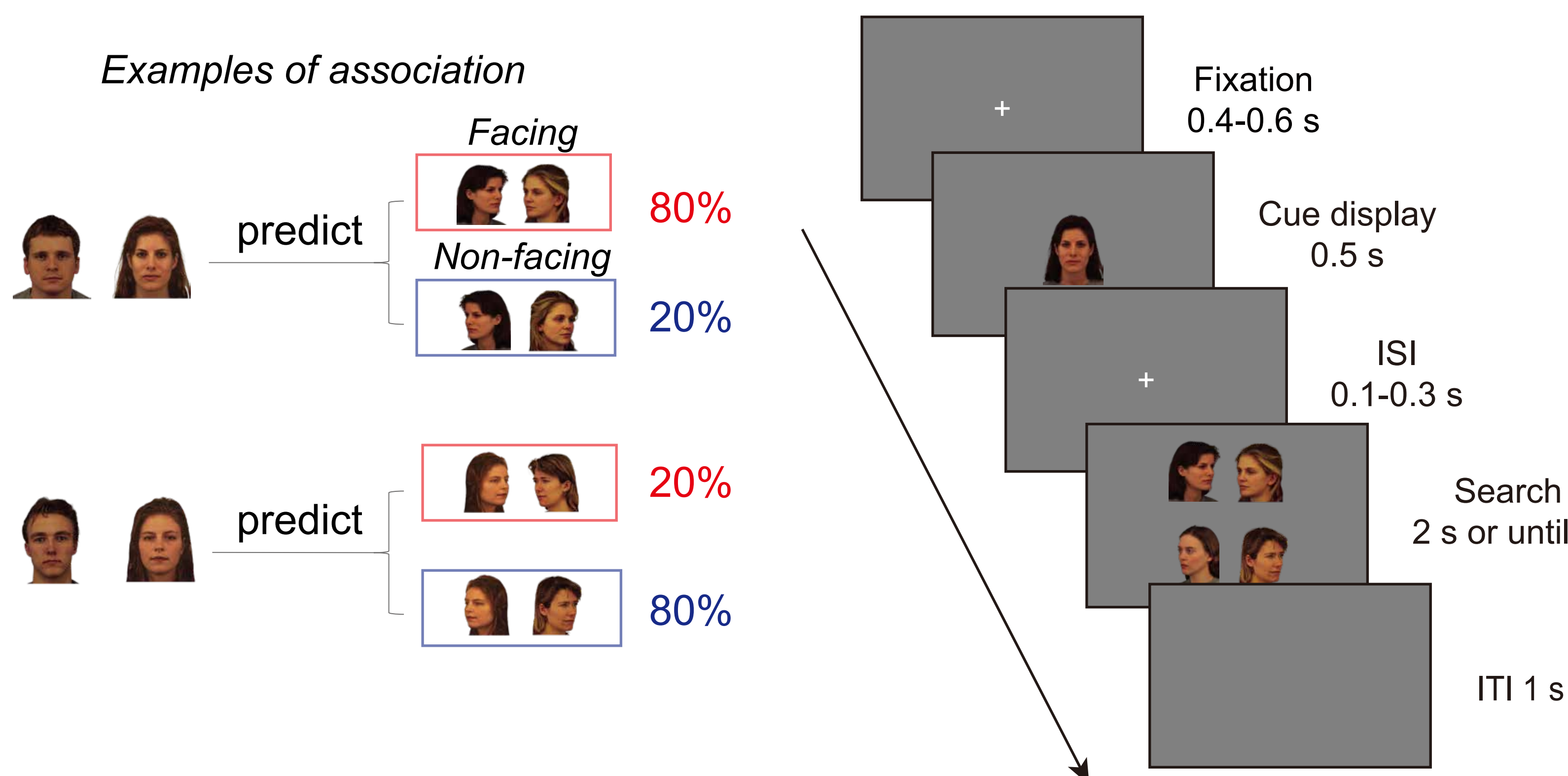
Introduction

- Face-to-face dyads are perceived as signaling ongoing social interaction and are prioritized by visual attention over non-facing dyads.
- However, attention and perception are shaped not only by current visual input but also by regularities acquired through past experience.

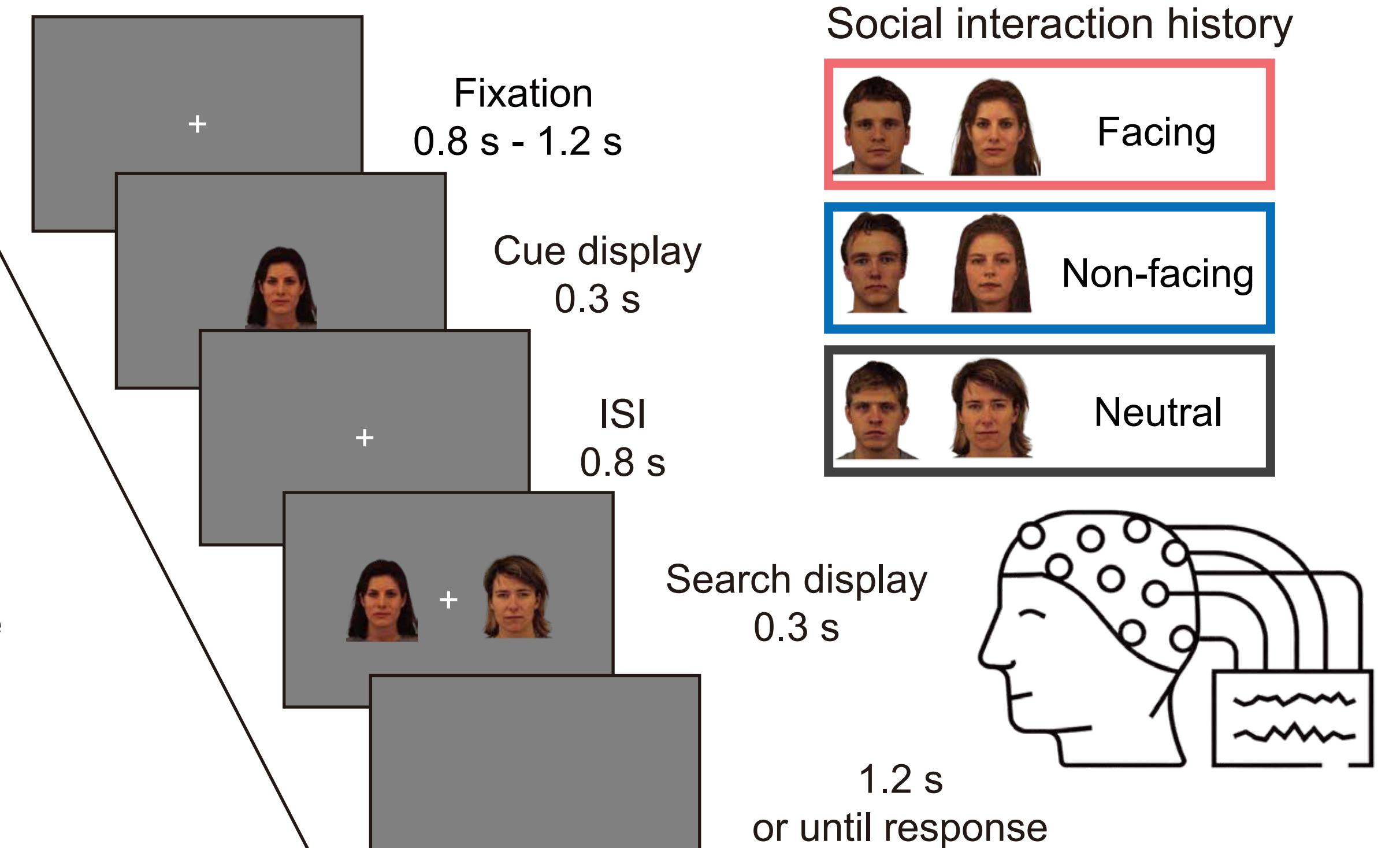
Does learning a face that usually appears in an interacting dyad alter face representations and attentional selection ?

Method

Day 1 · Associative learning

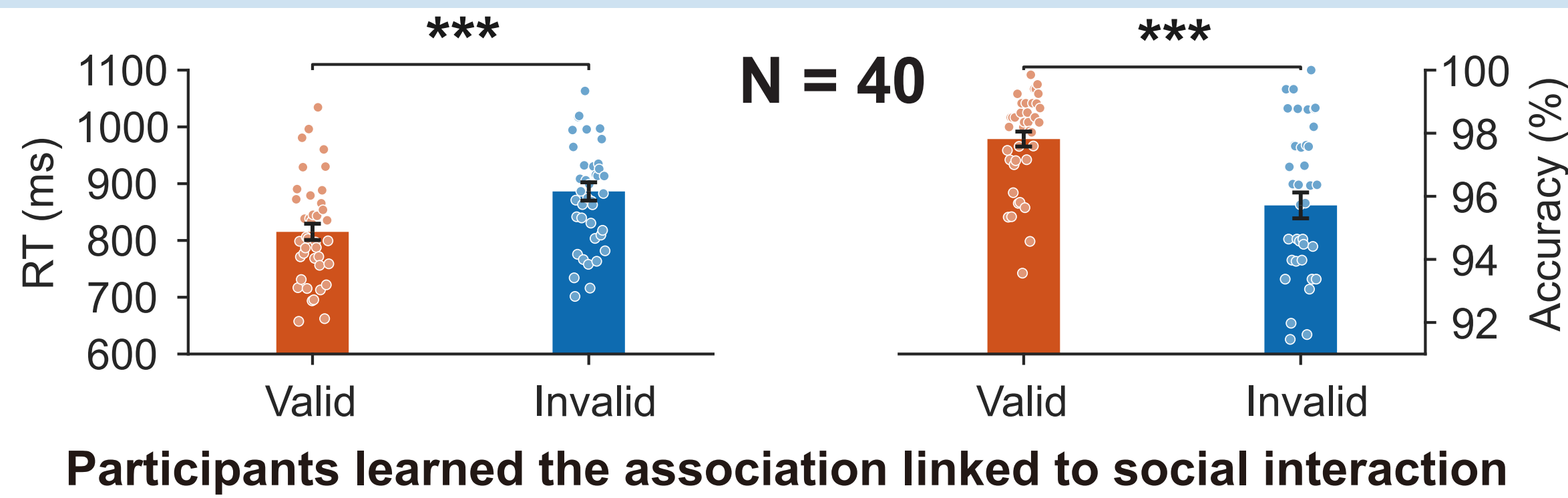


Day 2 · Test of social history effects (EEG)

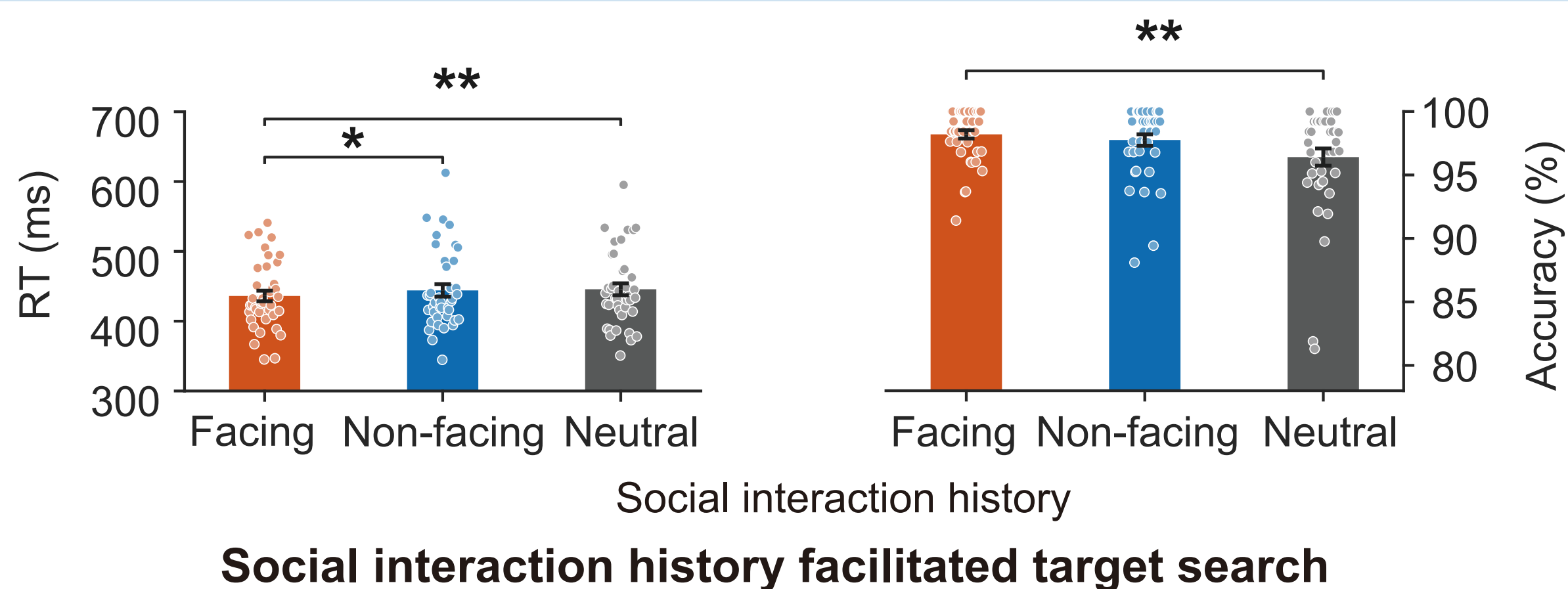


Results

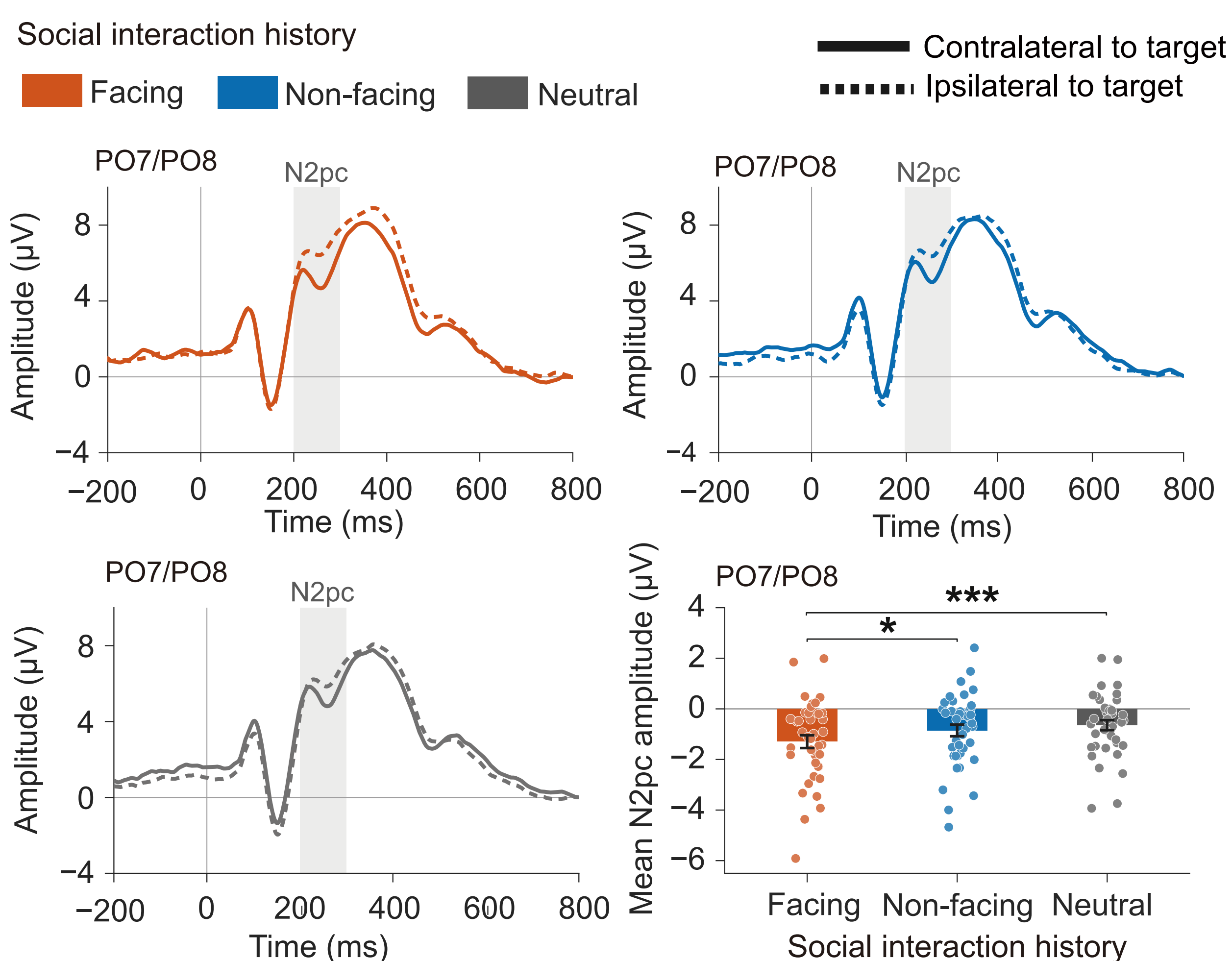
Training



Test

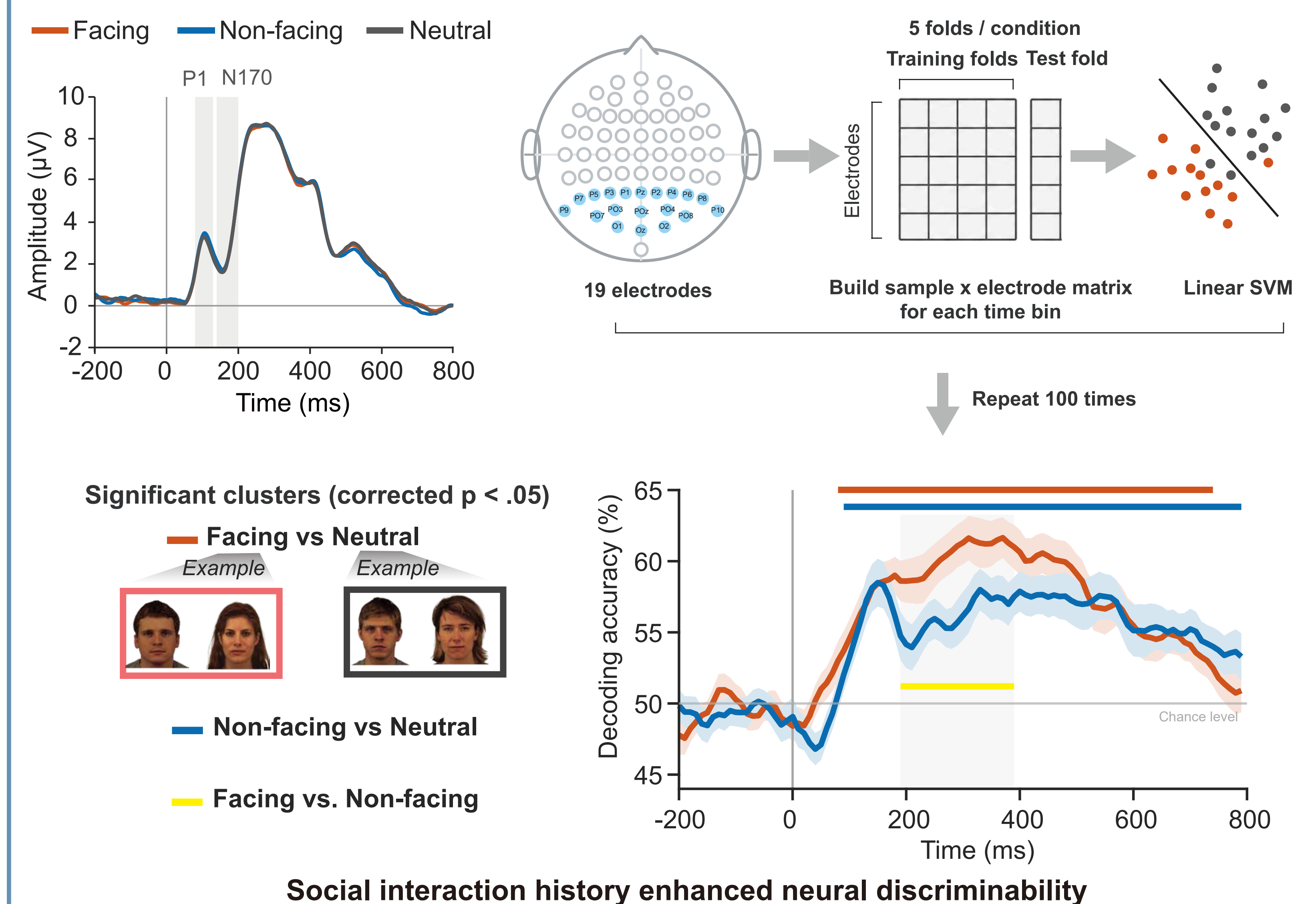


ERP analysis: Time-locked to search display

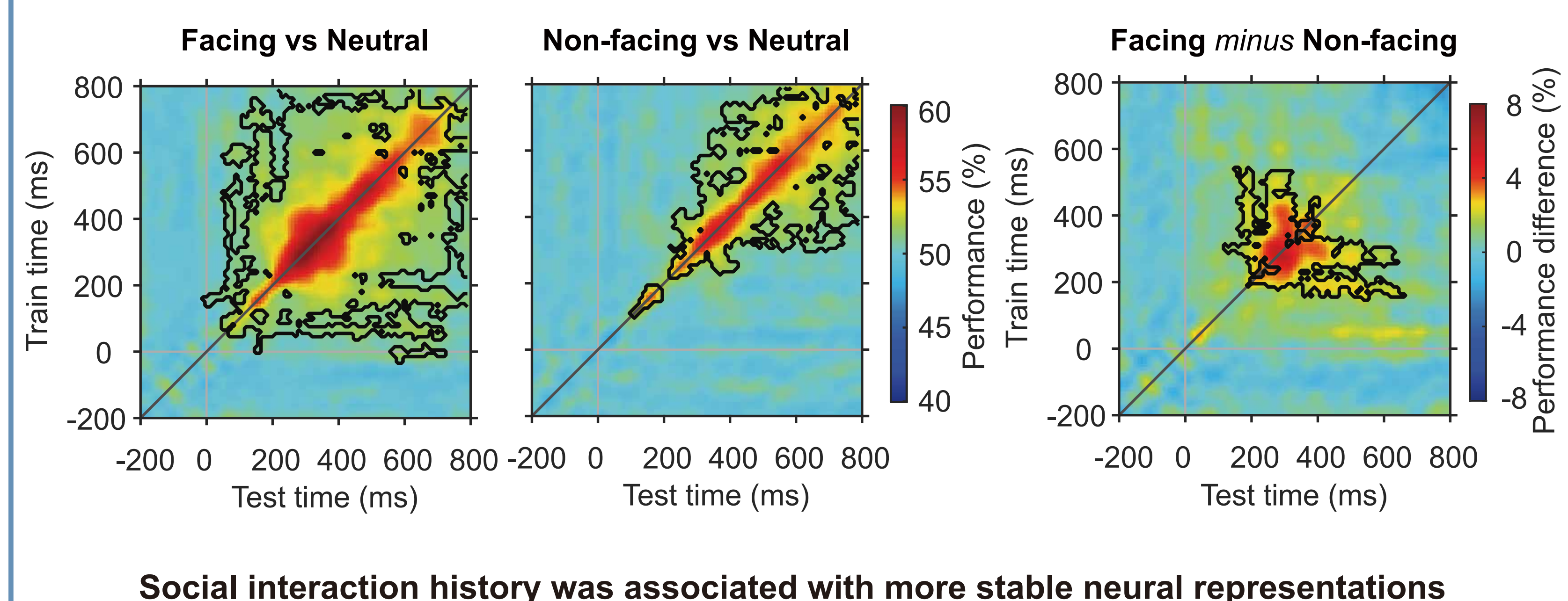


EEG decoding: Time-locked to cue display

ERP analysis: Time-locked to cue display



Temporal generalization



Conclusion

- Social interaction history enhanced the neural discriminability and temporal maintenance of face representations.
- The results suggest that learned social interaction is incorporated into face representations, shaping subsequent processing and selection.

If you have any questions about this study, please contact tanqs@zju.edu.cn